

Scale Up E-Commerce Performance Assistant

Final Compilation Report — Dynamic Operations Analysis in R

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1 Introduction

Modern e-commerce operations depend on fast, reliable fulfillment across thousands of orders and hundreds of sellers. An **e-commerce operations manager** must monitor payment economics, freight costs, delivery speed, seller risk, and data quality before changing staffing, SLAs, or audit priorities.

The **Scale Up E-Commerce Performance Assistant** addresses this need using a Level 2 Quarto statistical website built entirely in **R**. The tool supports decisions about:

- baseline delivery and payment performance,
- whether multi-seller coordination affects delivery time,
- which orders merit late-delivery screening,
- monthly volume and revenue trends,
- which sellers to review first,
- whether processed data are trustworthy enough for reporting.

The Olist Brazilian E-Commerce Public Dataset provides the empirical foundation. Because the reviews file was unavailable, review-score simulation was rejected and Skill 3 was redesigned around real **late-delivery** outcomes.

2 Dataset

The analysis uses six Olist tables joined at the order level after pre-aggregation to prevent revenue inflation from one-to-many relationships between orders, items, and payments.

Table	Grain	Role
Orders	One row per order	Status, purchase date, delivery timestamps
Customers	One row per customer	Geographic attributes
Order items	One row per line item	Price, freight, seller
Payments	One row per installment	Payment value and method

Table	Grain	Role
Products	One row per product	Catalog attributes
Sellers	One row per seller	Merchant identifiers

Joining strategy: item and payment tables are aggregated to one row per `order_id` before joining to orders. The processed order-level dataset contains **99,441** rows; the seller-order dataset contains seller-level records for prioritization.

Key processed fields include `total_payment_value`, `total_freight_value`, `actual_delivery_days`, `late_delivery`, `single_seller_order`, and `purchase_month`. Validation helpers in `R/validation_helpers.R` enforce variable existence, grouping rules, and logical consistency before each skill runs.

3 Explore Skill

3.1 Business question

What do order value, freight cost, delivery time, status mix, and late-delivery patterns look like across the platform?

3.2 Method

Descriptive statistics, missing-value counts, status tables, and distribution plots on the processed order-level dataset.

3.3 Results

Metric	Value
Total orders	99,441
Delivered/completed orders	96,478
Canceled or unavailable orders	1,234
Missing actual delivery dates	2,965
Median payment value	R\$ 105.29
Median freight value	R\$ 17.09
Median delivery time (completed)	10.22 days
Late deliveries	7,826
Late-delivery rate	8.11%

Payment values are right-skewed: the median (R\$ 105.29) is lower than the mean (R\$ 160.99), indicating a long tail of high-value orders. Among completed orders with valid delivery durations, **8.11%** arrived after the estimated delivery date.

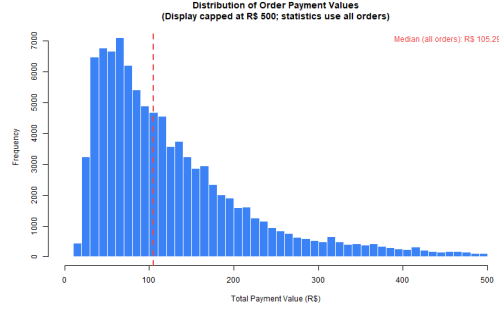


Figure 1: Distribution of order payment values.

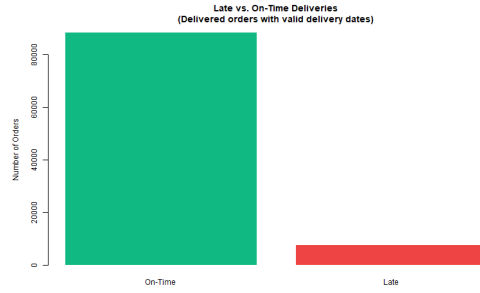


Figure 2: Late vs. on-time deliveries among completed orders.

3.4 Decision supported

These benchmarks help managers set realistic SLAs, freight expectations, and baseline late-delivery monitoring thresholds.

4 Compare Skill

4.1 Business question

Do multi-seller orders take longer to deliver than single-seller orders?

4.2 Method

Completed, non-canceled orders with valid delivery durations were split by `single_seller_order`. Because delivery times are skewed and contain outliers, a **Wilcoxon rank-sum test** was selected (Wilcoxon rank-sum test). The reported location shift uses the Hodges–Lehmann estimator (multi minus single).

4.3 Results

Group	N	Median delivery days
Single-seller	95,195	10.25
Multi-seller	1,275	7.93

- **Hodges–Lehmann location shift (multi — single):** -2.28 days
- **95% confidence interval:** [-2.61, -1.99] days
- **p-value:** 5.1e-50

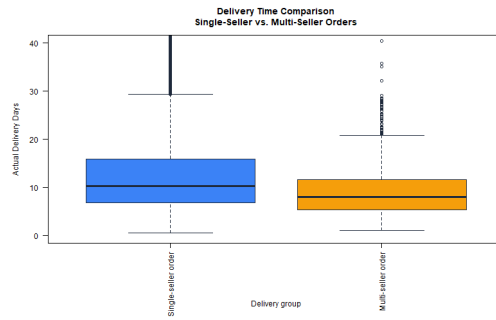


Figure 3: Delivery time comparison by seller structure.

4.4 Interpretation

Multi-seller orders have a **shorter** median delivery time (7.93 vs. 10.25 days). The negative location shift (-2.28 days) is statistically significant, but this is **observational** data. We cannot conclude that adding sellers *causes* faster delivery; unmeasured confounders such as product category, region, or carrier assignment may explain the association. Managers should treat this as a descriptive fulfillment comparison, not proof of coordination efficiency.

5 Model Skill

5.1 Business question

Which order characteristics are associated with late delivery among completed orders?

5.2 Method

Logistic regression on **96,469** completed orders with valid delivery timestamps and predictors. The data were split **77,175** training / **19,294** test with stratification on `late_delivery`. Late prevalence is **8.11%**. Because the minority class is rare, the classification threshold was selected on **training data only** using Youden’s J: **0.081**.

Monetary predictors use **log1p transforms** when skewness exceeds 1. Odds ratios describe multiplicative changes on the log-odds scale; they must **not** be read as one-real changes in raw payment or freight amounts.

5.3 Test-set performance

Metric	Value
Accuracy	0.672
Sensitivity	0.671
Specificity	0.672
Precision	0.153
F1 score	0.249
ROC AUC	0.721

	Term	Coefficient	p-value	Odds ratio	OR lower	OR upper
1	(Intercept)	-1.2403	0.0575	0.0804	0.0224	0.2893
31	seller_count	-1.7227	0.0000	0.1048	0.0615	0.1786
4	log1p_total_freight_value	0.7099	0.0000	1.9037	1.7819	2.0339
3	item_count	-0.4097	0.0000	0.6196	0.5782	0.6639
5	log1p_total_item_price	-0.0517	0.0045	0.9164	0.8843	0.9497
2	estimated_delivery_days	-0.0472	0.0000	0.9496	0.9454	0.9539
6	maximum_installments	0.0150	0.0112	1.0034	0.9919	1.0151

5.4 Interpretation

Accuracy near **0.672** is misleading because predicting the majority on-time class would achieve similar accuracy while catching almost no late orders. Precision is only **0.153**, so most flagged orders would still be on time. The model is therefore a **screening tool** for operational follow-up, not an automatic approve/reject rule. ROC AUC of **0.721** shows moderate discrimination. All coefficients represent **associations**, not causal effects.

6 Monthly Sales Trend Analysis

6.1 Business question

How did monthly completed-order volume and total payment value change from **2016-09** to **2018-08**?

6.2 Method

Monthly aggregation of completed orders with valid purchase month and payment value, plus month-over-month percentage changes. Boundary months with partial tracking windows (2016-09) and low-volume months are excluded from MoM comparisons.

6.3 Results

- Completed orders analyzed: **96,478**
- Total payment value: **R\$ 15,422,461.77**
- Peak complete month (orders): **2017-11** — **7,289** orders, **R\$ 1,153,528.05** payment value
- Largest order-volume increase: **2017-02** (120.4% MoM)
- Largest order-volume decrease: **2017-12** (-24.4% MoM)

	Month	Completed orders	Orders MoM %	Payment (R\$)	Payment MoM %
2	2016-10	265	26,400.00	46,566.71	NA
4	2017-01	750	74,900.00	127,545.67	649,979.87
5	2017-02	1,653	120.40	271,298.65	112.71
6	2017-03	2,546	54.02	414,369.39	52.74
7	2017-04	2,303	-9.54	390,952.18	-5.65
8	2017-05	3,546	53.97	567,066.73	45.05
9	2017-06	3,135	-11.59	490,225.60	-13.55
10	2017-07	3,872	23.51	566,403.93	15.54
11	2017-08	4,193	8.29	646,000.61	14.05
12	2017-09	4,150	-1.03	701,169.99	8.54
13	2017-10	4,478	7.90	751,140.27	7.13
14	2017-11	7,289	62.77	1,153,528.05	53.57
15	2017-12	5,513	-24.37	843,199.17	-26.90
16	2018-01	7,069	28.22	1,078,606.86	27.92
17	2018-02	6,555	-7.27	966,510.88	-10.39
18	2018-03	7,003	6.83	1,120,678.00	15.95
19	2018-04	6,798	-2.93	1,132,933.95	1.09
20	2018-05	6,749	-0.72	1,128,836.69	-0.36
21	2018-06	6,099	-9.63	1,012,090.68	-10.34
22	2018-07	6,159	0.98	1,027,903.86	1.56
23	2018-08	6,351	3.12	985,414.28	-4.13

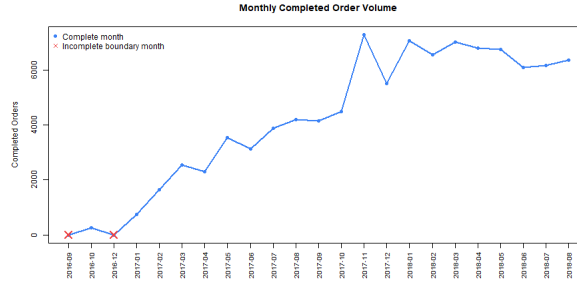


Figure 4: Monthly completed-order volume and payment trends.

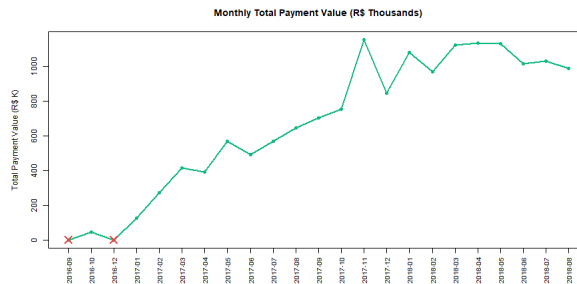


Figure 5: Monthly completed-order volume and payment trends.

6.4 Interpretation

Volume and revenue rose sharply during 2017, with **2017-11** as the busiest month. November 2017 coincides with Black Friday, which is a plausible operational explanation, but **seasonality is not statistically proven** here. Month **2016-11** is missing from the series and should be annotated in any dashboard. Managers can use these trends for staffing and revenue planning while treating boundary months cautiously.

7 Seller Prioritization

7.1 Business question

Which sellers should an operations manager review first?

7.2 Method

Of **3,095** sellers in the processed seller-order table, **804** met eligibility: at least **20 total seller-order records** and **20 valid delivered orders**. A weighted risk score ranks eligible sellers:

$$\text{priority_score} = 40\% * \text{late_delivery_rate_norm} + 25\% * \text{canceled_or_unavailable_rate_norm} + 20\% * \text{positive_median_delay_norm} + 15\% * \log1p(\text{order_volume})_norm$$

Higher scores indicate higher operational review priority.

7.3 Top 10 priority sellers

Rank	Seller ID	Seller-order records	Valid deliveries	Late rate	Cancel/un-avail. rate	Median delay (days)	Priority score
1	821fb029fc6e495ca4f08a35d51e53a5	5385	24	0.375	0.036	0	0.518
2	ede0c03645598cdfc63ca8237acbe43d	43	43	0.349	0.022	0	0.464
3	ad781527c93d00d89a11eecd9dca47c1	47	38	0.316	0.023	0	0.431
4	835f0f7810c76831d6c7d24c7a64644d	44	42	0.310	0.023	0	0.424
5	f76a3b1349b6df1ee875d1f3fa434240	40	24	0.375	0.000	0	0.406
6	66e0557ecc2b4dbea057e93f215f68d8	38	30	0.267	0.032	0	0.395
7	2a1348e9addc1af5aaa619b1a367926b	26	48	0.271	0.019	0	0.378
8	54965bbe3e4f07ae045b90b0b8547852	52	73	0.301	0.000	0	0.366
9	bc47d5d1490df2b36add65d733e2f4ba	24	21	0.095	0.083	0	0.357
10	06a2c3af7b3ace5d69171b0e14f03927	27	389	0.231	0.000	0	0.345

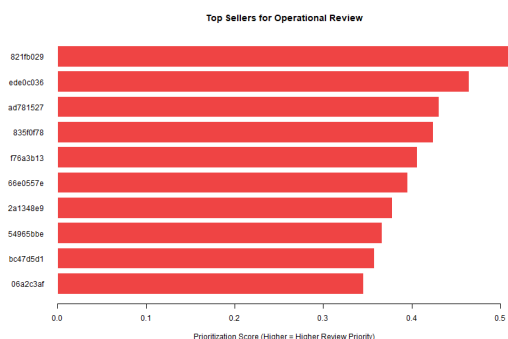


Figure 6: Top seller prioritization scores.

7.4 Interpretation

The highest-priority seller (821fb029fc6e495ca4f08a35d51e53a5) has a late-delivery rate of **37.50%** and priority score **0.518**. Because delivery outcomes are recorded at the **order level**, multi-seller orders assign the same delivery result to each participating seller. Seller rankings should therefore be interpreted as a prioritization aid, not a definitive blame assignment.

8 Data Quality Audit

8.1 Business question

Are missing values, duplicates, inconsistent dates, or payment mismatches affecting analysis?

8.2 Method

Programmatic checks on raw and processed tables with severity labels (High, Warning, Information).

8.3 Summary

- Total issues summarized: **27**
- High: **4**
- Warning: **8**
- Information: **15**
- Duplicate processed order keys: **0**
- Missing actual delivery dates: **2965** (~**2.98%** of orders)

	Issue	Count	Percent	Severity
10	zero_payment_values	4	0.004	Warning
11	missing_purchase_dates	0	0.000	Warning
12	missing_estimated_delivery_dates	0	0.000	Warning
14	actual_delivery_before_purchase	0	0.000	High
15	estimated_delivery_before_purchase	0	0.000	High
16	approval_before_purchase	0	0.000	High
17	delivered_without_actual_delivery_date	8	0.008	High
18	canceled_or_unavailable_with_actual_delivery_date	6	0.006	Warning
19	orders_without_item_records	775	0.779	Warning
20	orders_without_payment_records	1	0.001	Warning
25	missing_or_invalid_seller_count	775	0.779	Warning
27	missing_months_in_time_series	1	4.348	Warning

Notable findings include **775** orders without item records, **1** order without payment, **8** delivered orders without actual delivery date, **6** canceled/unavailable orders with delivery dates, **4** zero-payment orders, **1033** payment reconciliation differences beyond R\$0.05, **1278** multi-seller orders, and missing month **2016-11** in the time series.

8.4 Decision supported

Managers should resolve High-severity timestamp and delivered-status issues before trusting delivery KPIs, and treat payment mismatches as review items rather than automatic errors (vouchers and installments may explain differences).

9 Google Antigravity and Agent Skills

This project used **Google Antigravity** for initial scaffolding and **Agent Skills** (SKILL.md files) to standardize development. Later Cursor agents completed validation, statistical corrections, Quarto integration, and this final report.

Phase	Request	Output	Student review
Structure	Create Level 2 Quarto website skeleton	<code>_quarto.yml</code> , page scaffolds	Accepted layout; renamed Skill 3 after reviews file was missing
Data audit	Inspect CSV grains and join risks	<code>dataset.qmd</code> , import/clean drafts	Accepted pre-aggregation concept; added stronger validation later
Skill scaffolds	Create six <code>SKILL.md</code> files	<code>skills/*/SKILL.md</code>	Accepted format; rejected simulated review scores

Rejected redesign: Once `olist_order_reviews_dataset.csv` was confirmed unavailable, review-score simulation was rejected and the project was rebuilt around real late-delivery modeling.

10 Application or Delivery

Operations managers render the website from the project root:

```
& "C:\Scale Up_Project\Original working Scale Up AI Manager\bin\quarto.exe" render
```

The static site is published to `_site/index.html`. The final report renders separately:

```
& "C:\Scale Up_Project\Original working Scale Up AI Manager\bin\quarto.exe" render report/final
```

Each Quarto page documents one skill; processed CSV/RDS outputs in `output/` and `data/processed/` support audit trails. The tool is intended for local or intranet deployment as a decision-support dashboard, not autonomous order routing.

11 Limitations

1. **Dataset limitation:** The Olist reviews file is missing, historical coverage ends in 2018, and 2016–11 is absent from monthly trends. Approximately **2.98%** of orders lack actual delivery dates.
2. **Statistical limitation:** All comparisons and regression results are associative. Class imbalance makes accuracy misleading for late-delivery prediction, and seller metrics inherit shared order-level delivery outcomes for multi-seller orders.
3. **Implementation limitation:** Analyses use base R for stability; the website excludes the final report from default site render (`!report/final_report.qmd` in `_quarto.yml`) to keep PDF compilation separate. Plot regeneration depends on running analysis scripts during report render.

12 Reflection

This project reinforced how operational analytics depends as much on **data engineering** as on modeling. Pre-aggregating one-to-many tables, validating joins, and documenting exclusions prevented inflated revenue and false conclusions. The hardest steps were correcting seller eligibility rules, choosing robust nonparametric tests for skewed delivery times, and explaining why a statistically significant multi-seller difference does not imply causation.

The three additional skills—monthly trend analysis, seller prioritization, and data quality audit—were selected because an operations manager needs more than descriptive charts and one hypothesis test. Trends support staffing, prioritization converts seller performance into actionable review lists, and the audit builds trust before KPIs reach leadership.

With more time, I would add interactive filters by state or product category, externalize model thresholds by business cost, capture a website screenshot for documentation, and explore survival models for time-to-delivery rather than binary lateness only.